Notes: Malmendier & Nagel (QJE, 2016)
Learning from Inflation Experiences

Contents

1 Big picture 2
2 Data and measurement 3
2.1 Why inflation expectations are the right outcome 3
2.2 Inflation experience data 3
2.3 Michigan Survey of Consumers expectations data 3
2.4 Age groups used for visualization 3
2.5 SCF data for financial decisions 4
2.6 Aggregate expectations data 4
3 Models and empirical specifications 5
3.1 Learning-from-experience intuition 5
3.2 Perceived AR(1) law of motion 5
3.3 Recursive learning rule 5
3.4 Age-dependent gain 5
3.5 Experience-based forecast and common information 6
3.6 Baseline estimating equation 6
3.7 Alternative common-factor specifications 6
3.8 Financial-decision regression 6
3.9 Aggregate expectations and constant-gain learning 7
4 Identification and empirical design 8
4.1 What identifies the main expectations effect 8
4.2 Why age-dependent updating matters 8
4.3 Addressing age-specific inflation-basket confounds 8
4.4 Addressing imputation and sample concerns 8
4.5 Why the financial-decision tests matter 8
4.6 What the paper cannot fully prove 9
5 Tables and figures 10
5.1 Figure I + Figure IV: cohort disagreement and fitted expectations 10
5.2 Figure II + Figure VI: individual learning weights and aggregate constant gain 10
5.3 Figure III + Table I: inflation history and baseline estimates 11
5.4 Figure V + Figure VII: perceived inflation process and aggregate expectations 12
5.5 Table II + Table III: household financial decisions 13
5.6 Table IV: explaining mean inflation expectations 14
6 Short summary 15
7 Conclusion 15
7.1 Core conclusion 15
7.2 Economic intuition 15
7.3 Takeaway 15

1 Big picture

- **Question.** How do households form expectations about future inflation, and why do different generations disagree so much?
- **Core mechanism.** People overweight inflation realizations experienced during their lifetimes. Young individuals update more strongly because recent data account for a larger share of their lifetime history.
- **Main model.** Individuals recursively estimate a perceived AR(1) law of motion for inflation using lifetime inflation data. The gain is decreasing in age rather than in calendar time.
- **Empirical setting.** The paper uses 57 years of Reuters/Michigan Survey of Consumers microdata on one-year inflation expectations, matched to long historical CPI inflation data.
- **Identification idea.** Time dummies absorb any common macro news or professional forecasts. The key estimate comes from cross-sectional differences in expectations across cohorts with different lifetime histories.
- **Main expectations result.** Cohort-specific experienced inflation strongly predicts cohort-specific inflation expectations. In the baseline, the gain parameter is $\theta = 3.044$, and the sensitivity of survey expectations to experience-based forecasts is about 0.672.
- **1970s interpretation.** Young people in the late 1970s and early 1980s expected higher inflation than older people because their lived history was dominated by high and persistent inflation.
- **Behavioral result.** Experience-induced inflation beliefs help explain real household financial choices: households with higher experience-based inflation expectations take on more fixed-rate nominal mortgage debt and hold less long-term nominal bonds.
- **Aggregate implication.** When averaged across cohorts, learning from experience closely resembles constant-gain adaptive learning. It gives a microfoundation for why aggregate expectations behave as if old data are discounted.
- **Economic takeaway.** Inflation expectations are not only a function of current information or rational forecasts. Personal macroeconomic history shapes beliefs and financial decisions, and generational turnover can move aggregate expectations.

2 Data and measurement

2.1 Why inflation expectations are the right outcome

- Inflation expectations affect perceived real interest rates, household borrowing, asset allocation, expenditure decisions, and monetary-policy transmission.
- The paper focuses on heterogeneity in expectations, especially age-based disagreement that standard aggregate adaptive-learning models cannot explain.
- The Reuters/Michigan Survey of Consumers is useful because it provides long-run microdata rather than only aggregate means or medians.

Interpretation. The key empirical object is not simply whether inflation expectations follow past inflation. The central question is whether people with different lifetime experiences have different forecasts at the same point in time.

2.2 Inflation experience data

- Inflation experiences are measured using annualized quarterly log CPI inflation rates.
- The long inflation series runs from 1872 to 2009 and is taken from Robert Shiller's historical data.
- The long history matters because older respondents in the early Michigan surveys had lived through inflation realizations long before the survey period.
- The model treats each birth cohort as having accumulated a lifetime data set of inflation realizations.

Interpretation. The experienced-inflation variable is cohort-specific. A person aged 30 and a person aged 70 in the same quarter observe the same current inflation, but their lifetime data sets are very different.

2.3 Michigan Survey of Consumers expectations data

- The Michigan Survey has inflation expectations data from 1953 to 2009.
- The survey initially occurred three times a year, then quarterly from 1960 to 1977, and monthly from 1978 onward.
- The main survey outcome is the respondent's one-year-ahead inflation expectation.
- The authors focus on household heads aged 25 to 74, to maintain reasonable cohort sizes.
- The paper aggregates responses to the birth-year cohort level.
- For quarters with only categorical responses on expected price changes, the authors impute percentage expectations.

Interpretation. Cohort aggregation reduces noise from outlier forecasts and digit preference. It also directly matches the theory, which predicts disagreement across birth cohorts.

2.4 Age groups used for visualization

The paper often plots three age categories:

$$\text{young: age} < 40, \qquad \text{middle-aged: } 40 \leq \text{age} \leq 60, \qquad \text{old: age} > 60.$$

- These plots are not the core estimation sample; they are visual summaries of cohort-level results.
- The regression estimation uses cohort-level data and time dummies.

- Four-quarter moving averages are used in several figures to make lower-frequency age patterns easier to see.

Interpretation. The simple age-group figures show the phenomenon before the formal model: belief gaps can reach almost 3 percentage points in volatile inflation periods.

2.5 SCF data for financial decisions

- To test whether experience-based beliefs affect behavior, the paper uses the Survey of Consumer Finances.
- The SCF sample covers 19 surveys from 1960 to 2007.
- The data are again aggregated to birth-year cohort level.
- Key outcomes are per-capita fixed-rate mortgage balances, long-term bond holdings, new fixed-rate mortgages, new variable-rate mortgages, income, and net worth.
- Fixed-rate mortgages are the main nominal fixed-rate liability. Long-term bonds are the main nominal fixed-rate asset.

Interpretation. If households expect higher inflation, they perceive lower real borrowing costs on fixed nominal debt and lower real returns on fixed nominal assets. The model therefore predicts more fixed-rate borrowing and less long-term bond holding.

2.6 Aggregate expectations data

- The paper also studies aggregate expectations by averaging experience-based forecasts across cohorts.
- The aggregate comparison includes average Michigan survey expectations, Survey of Professional Forecasters forecasts, and sticky-information forecasts.
- This part asks whether a cross-sectionally estimated experience model can also match the time series of average expectations.

Interpretation. This is a demanding out-of-sample exercise because the baseline estimates use cross-sectional cohort differences, not the aggregate level of expectations.

3 Models and empirical specifications

3.1 Learning-from-experience intuition

The mechanism can be summarized as:

lifetime inflation history$_{s,t}$ $\Rightarrow$ cohort-specific perceived mean and persistence $\Rightarrow$ cohort-specific inflation forecast.

Interpretation. Young cohorts have a shorter lifetime history, so a recent inflation surprise has a larger effect on their perceived inflation process.

3.2 Perceived AR(1) law of motion

Individuals estimate a perceived AR(1) model for quarterly inflation:

$$\pi_{t+1} = \alpha + \phi\pi_t + \varepsilon_{t+1}. \tag{1}$$

Let

$$b_{t,s} = (\alpha_{t,s}, \phi_{t,s})' \quad \text{and} \quad x_t = (1, \pi_t)'.$$

Interpretation. The perceived mean and persistence can differ by cohort s, because each cohort weights lifetime inflation data differently.

3.3 Recursive learning rule

The cohort-specific recursive least-squares update is:

$$b_{t,s} = b_{t-1,s} + \gamma_{t,s} R_{t,s}^{-1} x_{t-1} \left(\pi_t - b'_{t-1,s} x_{t-1}\right), \tag{2}$$

$$R_{t,s} = R_{t-1,s} + \gamma_{t,s} \left(x_{t-1} x'_{t-1} - R_{t-1,s}\right). \tag{3}$$

Interpretation. The forecast changes when realized inflation differs from the model prediction. The strength of the update is governed by the gain $\gamma_{t,s}$.

3.4 Age-dependent gain

The key modification relative to standard adaptive learning is:

$$\gamma_{t,s} = \begin{cases} \dfrac{\theta}{t-s}, & t-s \geq \theta, \\ 1, & t-s < \theta. \end{cases} \tag{4}$$

- $t - s$ is the cohort's age measured in quarters.
- θ controls how quickly old experiences are downweighted.
- If $\theta = 1$, all observations since birth are equal-weighted.
- If $\theta > 1$, recent observations receive more weight than early-life observations.

Interpretation. The gain is decreasing in age, not in calendar time. Therefore two people in the same quarter update differently after the same inflation surprise.

3.5 Experience-based forecast and common information

Let $\hat{\pi}_{t+1|t,s}$ denote the one-year-ahead forecast generated by cohort s's learned AR(1) model. The paper allows expectations to also depend on common information:

$$\pi^e_{t+1|t,s} = \beta\hat{\pi}_{t+1|t,s} + (1-\beta)f_t. \tag{5}$$

Interpretation. The model does not assume people use only lifetime experiences. It estimates the incremental effect of lifetime experiences after allowing for common factors such as professional forecasts or media information.

3.6 Baseline estimating equation

The main estimating equation is:

$$\widetilde{\pi}_{t+1|t,s} = \beta\hat{\pi}_{t+1|t,s}(\theta) + \lambda' D_t + \varepsilon_{t,s}. \tag{6}$$

- $\widetilde{\pi}_{t+1|t,s}$ is the measured Michigan survey expectation for cohort s at time t.
- D_t are time dummies.
- $\hat{\pi}_{t+1|t,s}(\theta)$ is nonlinear in the gain parameter θ.
- The parameters θ and β are estimated by nonlinear least squares.
- Standard errors are two-way clustered by cohort and quarter.

Interpretation. Time dummies absorb any aggregate information common to all cohorts. Identification comes from the evolution of cross-sectional belief gaps across cohorts.

3.7 Alternative common-factor specifications

The paper also replaces time dummies with observable common factors:

$$\pi^e_{t+1|t,s} = \beta\hat{\pi}_{t+1|t,s} + (1-\beta)SPF_{t-1} + (1-\beta)\zeta_t, \tag{7}$$

or

$$\pi^e_{t+1|t,s} = \beta\hat{\pi}_{t+1|t,s} + (1-\beta)\bar{\hat{\pi}}_{t+1|t} + (1-\beta)\zeta_t. \tag{8}$$

Interpretation. The Survey of Professional Forecasters and the average experience-based forecast both work as plausible common components. The key β estimate remains similar.

3.8 Financial-decision regression

To test whether inflation experiences affect household balance sheets, the paper estimates:

$$y_{t,s} = \eta_1\hat{\pi}_{t+1|t,s} + \eta_2' X_{t,s} + \eta_3' A_{t,s} + \eta_4' D_t + \xi_{t,s}. \tag{9}$$

- $y_{t,s}$ is a cohort-level measure of fixed-rate liabilities or fixed-rate assets.
- $X_{t,s}$ includes income and net worth controls.
- $A_{t,s}$ are age dummies.
- D_t are survey-year dummies.

Interpretation.A higher experience-based inflation forecast should increase fixed-rate borrowing and decrease fixed-rate nominal asset holding.

3.9 Aggregate expectations and constant-gain learning

Averaging across cohorts gives:

$$\bar{\pi}^e_{t+1|t} = \beta \bar{\hat{\pi}}_{t+1|t} + (1-\beta)\zeta_t. \tag{10}$$

The paper shows that $\bar{\hat{\pi}}_{t+1|t}$ is close to a constant-gain forecast with gain approximately:

$$\gamma = 0.0180.$$

Interpretation. Generational replacement creates aggregate downweighting of distant history. This gives a microfoundation for constant-gain learning.

4 Identification and empirical design

4.1 What identifies the main expectations effect

- The main specification includes quarter fixed effects.
- These fixed effects remove common information available to all cohorts at the same time.
- The remaining variation is cross-sectional: different cohorts have different lifetime histories of inflation.
- The model is rejected if cohort differences in experienced inflation do not predict cohort differences in expectations.

Interpretation. This is the crucial design advantage over aggregate adaptive-learning regressions. Aggregate regressions can confuse adaptive learning with other macro mechanisms; cross-cohort variation directly tests the experience channel.

4.2 Why age-dependent updating matters

- A young person has only a short observed history, so one inflation surprise is important.
- An older person has a longer history, so the same surprise is diluted by many past observations.
- The model therefore predicts that belief disagreement is largest after volatile macroeconomic episodes.

Interpretation. The late 1970s and early 1980s are the central historical episode: young cohorts overweight high and persistent inflation, while older cohorts also remember low-inflation decades.

4.3 Addressing age-specific inflation-basket confounds

- Older and younger households may consume different bundles.
- The paper checks whether age-specific inflation differences explain the expectation gaps.
- It uses the experimental CPI for the elderly and a gasoline price series.
- These controls do not materially change the learning-rule estimates.

Interpretation. The results are not simply that old and young households face different current consumption-basket inflation rates.

4.4 Addressing imputation and sample concerns

- The baseline includes imputed percentage expectations for quarters with only categorical responses.
- Excluding imputed observations produces very similar estimates.
- The paper also checks alternative starting points for experience accumulation, including before birth and after birth.
- The fitted weight profiles remain close to the baseline.

Interpretation. The estimates are robust to data construction choices that could otherwise drive cohort differences.

4.5 Why the financial-decision tests matter

- Beliefs are economically meaningful only if they affect choices.
- Higher expected inflation lowers perceived real rates on fixed nominal debt.

- Therefore households with higher experience-based inflation forecasts should prefer fixed-rate liabilities and avoid long-term fixed nominal assets.
- The SCF results show exactly this pattern, especially for fixed-rate mortgages.

Interpretation. The mortgage and bond tests help validate the beliefs channel. Inflation experiences predict not just stated expectations but also balance-sheet positions.

4.6 What the paper cannot fully prove

- Lifetime experience is not randomly assigned.
- Age may correlate with other social, cognitive, or institutional differences.
- Survey expectations contain measurement error and digit preference.
- The financial-decision regressions are cohort-level and cannot fully capture all household-level constraints.
- The theory does not pin down whether experience works through memory, salience, availability, parental transmission, media attention, or social learning.

Interpretation. The evidence is strongest because many pieces line up: cohort expectations, estimated weights, perceived mean and persistence, household financial decisions, and aggregate expectations.

5 Tables and figures

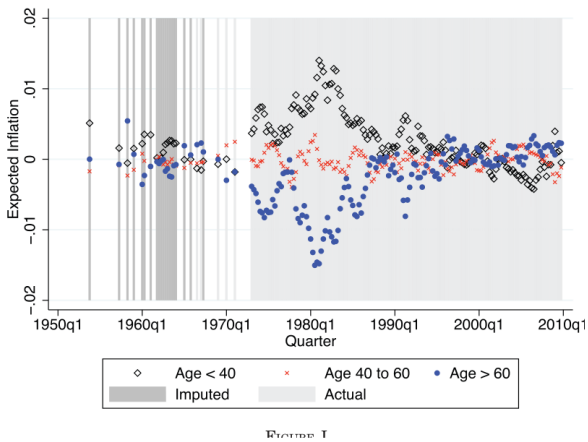
This section keeps adjacent visuals together to save space and make comparisons easier. All images are directly cropped from the original paper.

5.1 Figure I + Figure IV: cohort disagreement and fitted expectations

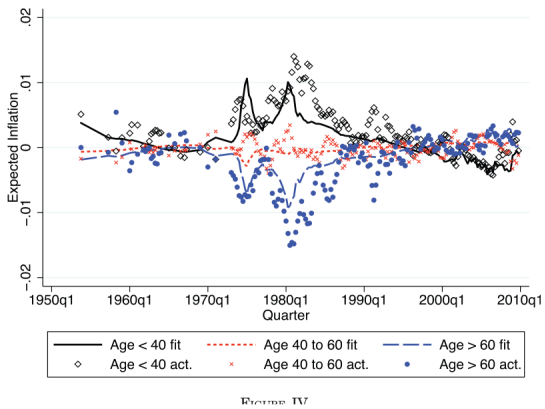
Read:

- Figure I shows age-group inflation expectations relative to the cross-sectional mean.
- Disagreement is especially large during and after the high-inflation 1970s and early 1980s.
- Young individuals expect higher inflation after the 1970s, while older individuals expect lower inflation.
- Figure IV overlays fitted values from the learning-from-experience model.
- The model captures the large young-old gap and the later reversals in the gap.

Economic message. The age pattern is not a small demographic detail. It is the main empirical signature of learning from lifetime macroeconomic experience.



(a) Figure I: inflation expectations by age group.



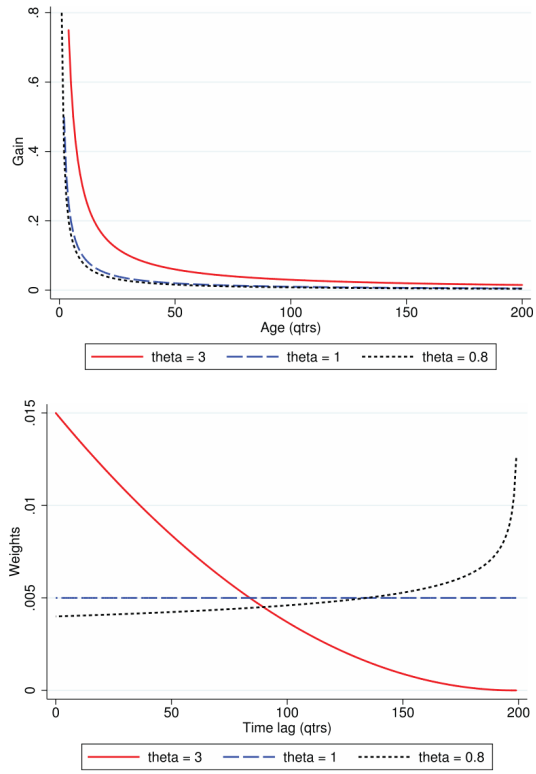
(b) Figure IV: actual and fitted expectations.

5.2 Figure II + Figure VI: individual learning weights and aggregate constant gain

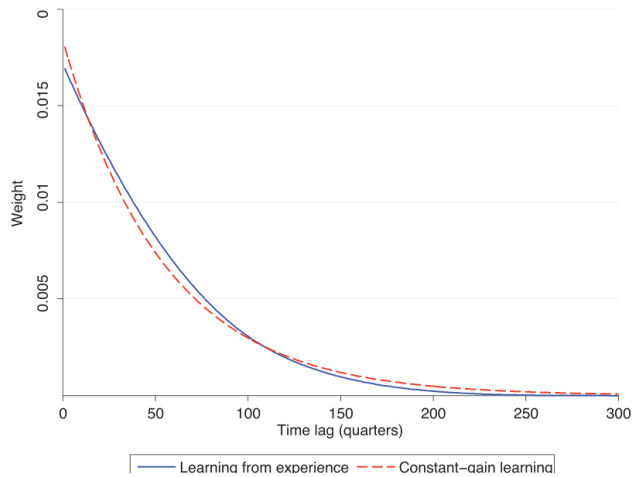
Read:

- Figure II shows the age-dependent gain for different values of θ.
- Higher θ means a stronger reaction to recent inflation and faster downweighting of old data.
- With $\theta = 3$, recent observations receive much more weight than early-life observations.
- Figure VI averages the experience-based weights across ages and compares them with constant-gain learning.
- The two weight profiles are almost the same when the constant gain is $\gamma = 0.0180$.

Economic message. At the individual level, learning is age-dependent. At the aggregate level, generational averaging makes the economy look as if it uses a constant-gain adaptive-learning rule.



(a) Figure II: gains and implied weights.



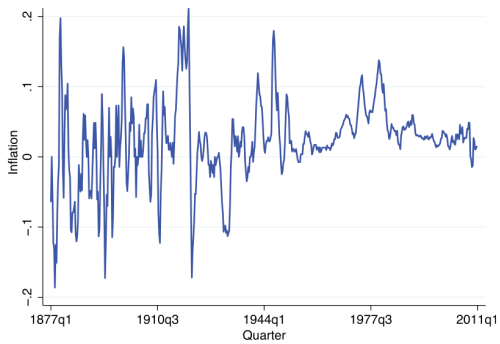
(b) Figure VI: mean weights vs. constant-gain learning.

5.3 Figure III + Table I: inflation history and baseline estimates

Read:

- Figure III shows long-run CPI inflation back to the late nineteenth century.
- The historical series includes very different inflation regimes, so cohorts born at different dates accumulate different histories.
- Table I reports the baseline estimates of the learning-from-experience model.
- The baseline gain parameter is $\theta = 3.044$ with standard error 0.233.
- The baseline sensitivity is $\beta = 0.672$ with standard error 0.076.
- Excluding imputed data barely changes the estimates: $\theta = 3.144$ and $\beta = 0.675$.
- Specifications using SPF or the average learning-from-experience forecast as the common factor produce similar sensitivity estimates.

Economic message. The baseline estimates show that lifetime experiences have a large incremental effect on expectations even after absorbing common time effects.



(a) Figure III: annual CPI inflation rates.

TABLE I

Learning-from-Experience Model: Estimates from Cohort Data

	(1)	(2)	(3)	(4)	(5)
	Baseline			Restricted	
Gain parameter θ	3.044	3.144	3.976	4.144	3.044
	(0.233)	(0.257)	(0.494)	(0.446)	—
Sensitivity β	0.672	0.675	0.664	0.693	0.672
	(0.076)	(0.079)	(0.084)	(0.111)	(0.093)
Time dummies	Yes	Yes	No	No	No
Imputed data included	Yes	No	No	No	No
Restrictions			$f_t = SPF_{t-1}$	$f_t = \bar{\tau}_{t+1\|t}$	$f_t = \bar{\tau}_{t+1\|t}$ $\theta = 3.044$
Adj. R^2	0.637	0.635	—	—	—
RMSE	0.0148	0.0152	0.0189	0.0191	0.0194
# Obs.	8,215	7,650	7,400	7,650	7,650

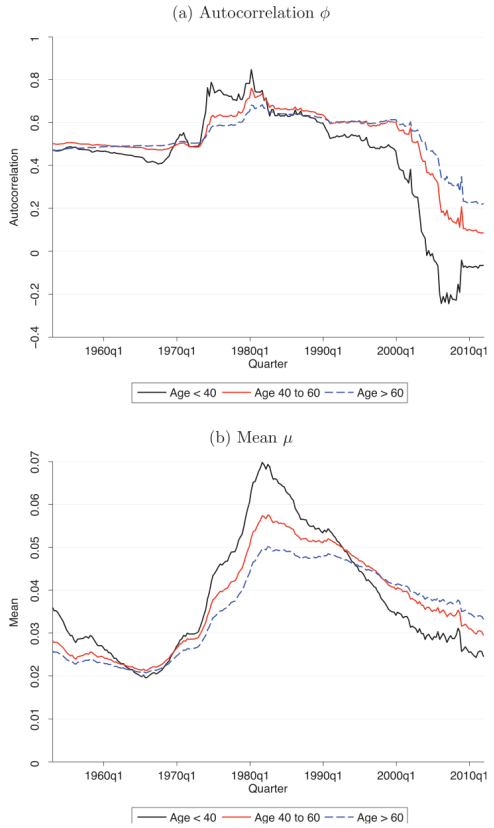
(b) Table I: learning-from-experience model estimates.

5.4 Figure V + Figure VII: perceived inflation process and aggregate expectations

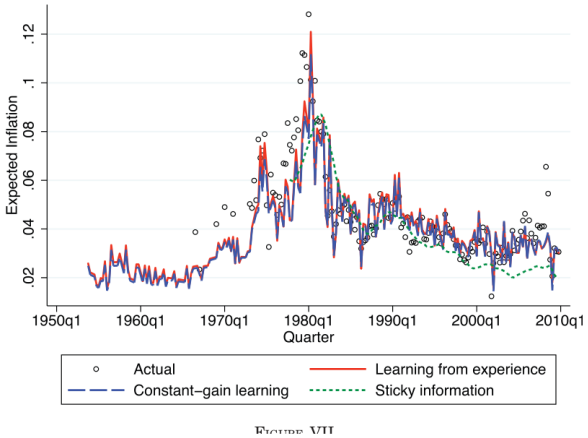
Read:

- Figure V shows cohort-specific perceived AR(1) parameters.
- Young individuals have more volatile perceived inflation means and autocorrelations.
- Around 1980, young individuals perceive both a higher mean and higher persistence of inflation.
- By the late 2000s, young individuals' perceived persistence is close to zero, indicating better anchoring.
- Figure VII shows that average experience-based forecasts closely track average survey expectations.
- Constant-gain forecasts are nearly indistinguishable from average experience-based forecasts.

Economic message. The model explains both cross-sectional disagreement and aggregate expectation dynamics. It also links improved anchoring to the experiences of younger cohorts.



(a) Figure V: perceived persistence and mean inflation.



(b) Figure VII: aggregate expectations and forecasts.

5.5 Table II + Table III: household financial decisions

Read:

- Table II summarizes SCF cohort-level variables.
- Mean log fixed-rate mortgage balances are 9.76; mean log long-term bond holdings are 8.61.
- New fixed-rate and variable-rate mortgage variables are available from 1983 onward.
- Table III relates household nominal positions to experience-based inflation forecasts.
- Fixed-rate mortgage debt has a large positive coefficient: 35.27 in the full sample.
- Long-term bond holdings have a negative coefficient: -20.56, though it is not statistically significant.
- New fixed-rate mortgages have a very large positive coefficient: 132.71.
- New variable-rate mortgages have a negative but statistically imprecise coefficient: -42.82.

Economic message. Households with higher experience-induced inflation expectations tilt toward fixed nominal liabilities. This is exactly the real-interest-rate channel predicted by the theory.

TABLE II

Survey of Consumer Finances: Summary Statistics of Cohort Aggregates

	(1) Log fixed-rate mortgages	(2) Log long-term bonds	(3) Log new fixed-rate mortgages	(4) Log new variable-rate mortgages	(5) Log income	(6) Log net worth
Mean	9.76	8.61	6.71	3.39	10.91	11.63
Std. dev.	1.29	1.85	3.45	3.72	0.41	0.89
p10	8.43	6.26	0.00	0.00	10.42	10.44
Median	10.05	8.73	8.21	0.00	10.93	11.62
p90	10.92	10.87	9.50	8.30	11.42	12.79
Sample	Full	Full	≥1983	≥1983	Full	Full

Notes. The SCF sample includes 19 surveys during the period from 1960 to 2007, and 18 of those have information on holdings of long-term bonds. The data on borrowing and bond holdings is aggregated to per capita numbers at the cohort level.

(a) Table II: SCF summary statistics.

TABLE III

Inflation Experiences and Household Nominal Positions

	(1) Fixed-rate mortgages	(2) Long-term bonds	(3) Fixed-rate mortgages	(4) Long-term bonds	(5) New fixed-rate mortgages	(6) New variable-rate mortgages
Learn.-from-exp. forecast	35.27	−20.56	26.77	−9.07	132.71	−42.82
	(8.39)	(13.74)	(4.47)	(6.92)	(25.08)	(55.57)
Log income	0.92	0.45	0.60	0.02	1.23	2.60
	(0.16)	(0.25)	(0.13)	(0.13)	(1.19)	(1.29)
Log net worth	−0.10	1.09	0.18	1.18	−0.56	−1.79
	(0.15)	(0.13)	(0.06)	(0.10)	(0.69)	(0.94)
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Full	Full	≥1983	≥1983	≥1983	≥1983
Adj. R^2	0.617	0.852	0.856	0.915	0.485	0.243
# Obs.	950	900	450	450	450	450

(b) Table III: inflation experiences and nominal positions.

5.6 Table IV: explaining mean inflation expectations

Read:

- Column (1) regresses average survey expectations on average learning-from-experience forecasts.
- The coefficient is 0.887 with adjusted $R^2 = 0.564$.
- Column (2) uses constant-gain learning and obtains a very similar coefficient of 0.931.
- Column (3) uses sticky information and obtains a coefficient of 0.878 with adjusted $R^2 = 0.602$.
- Column (4) includes both learning-from-experience and sticky-information forecasts.
- The learning-from-experience coefficient remains large, 0.695, and the adjusted R^2 rises to 0.715.

Economic message. Learning from experience is not just a sticky-information proxy. It adds explanatory power for average expectations even when sticky information is included.

TABLE IV

Explaining Mean Inflation Expectations

	(1)	(2)	(3)	(4)
Learning-from-experience forecast	0.887			0.695
	(0.120)			(0.132)
Constant-gain-learning forecast		0.931		
		(0.137)		
Sticky-information forecast			0.878	0.390
			(0.199)	(0.150)
Intercept	0.009	0.008	0.011	−0.000
	(0.005)	(0.005)	(0.006)	(0.005)
Adj. R^2	0.564	0.555	0.602	0.715
# Obs.	173	173	129	129

Table IV: Explaining mean inflation expectations

6 Short summary

- The paper asks how individuals form inflation expectations.
- It argues that people overweight inflation experienced during their own lives.
- The model modifies adaptive learning by making the gain depend on age.
- Younger individuals update more strongly because their lifetime data set is shorter.
- Individuals estimate a perceived AR(1) model of inflation from their experienced data.
- The model predicts time-varying disagreement across generations.
- Michigan Survey microdata from 1953 to 2009 show exactly this age-based disagreement.
- The baseline estimate $\theta = 3.044$ implies strong weight on recent lifetime experiences.
- The sensitivity estimate $\beta = 0.672$ means experience-based forecast differences strongly map into survey expectation differences.
- The model explains why young individuals expected much higher inflation after the 1970s.
- It also explains later reversals in the young-old expectation gap.
- The results are robust to excluding imputed survey observations.
- Age-specific consumption baskets and gasoline-price sensitivity do not explain the findings.
- Experience-based inflation expectations also predict financial behavior.
- Households with higher experience-based inflation forecasts hold more fixed-rate mortgage debt.
- They hold less long-term nominal bonds, though the bond result is less precisely estimated.
- For new mortgages after 1983, higher experience-based expected inflation predicts more fixed-rate borrowing.
- Averaging across cohorts makes the experience model look like constant-gain learning.
- The model matches aggregate survey expectations despite being estimated from cross-sectional cohort differences.
- The broad lesson is that macroeconomic expectations have personal-history foundations.

7 Conclusion

7.1 Core conclusion

Malmendier and Nagel show that inflation expectations are shaped by the inflation histories people personally experience. Different cohorts have different perceived means and persistence of inflation, and these differences explain substantial disagreement in survey expectations.

7.2 Economic intuition

Recent inflation matters more for young people because it is a larger fraction of their lifetime data. Older people have longer histories, so the same inflation shock is diluted by earlier experience. At the aggregate level, generational turnover makes society collectively discount distant inflation history.

7.3 Takeaway

The paper gives behavioral macroeconomics a disciplined empirical foundation: lived macroeconomic experience affects beliefs and decisions. Inflation expectations are therefore partly historical, cohort-specific, and path-dependent, which matters for monetary policy and household finance.